॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

PROJECT REPORT

Submitted for

HACK-O-WEEK PROGRAM

"TechMart — Shopping Cart Using Linked List"

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

Submitted By

1. **Swanandi Thakre – 25070521086**
2. **Girija Mahajan - 25070521087**
3. **Vansh Gaikwad - 25070521088**

UNDER THE GUIDANCE OF

Prof. Piyush Chauhan

Designation
Associate professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING A Y 2025-26

CERTIFICATE

This is to certify that the project entitled ***"TechMart — Shopping Cart Using Linked List"*** is carried out by the **Swanandi Thakre, Girija Mahajan, Vansh Gaikwad** in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2025-2026.

Name of Mentor --**Prof. Piyush Chauhan**

Designation-- **Associate professor**

Dr. Gaurav Londhe

Associate Professor

Hack-O-Week In-Charge

Dr. Sagar Kumar Badhiye

HoD, Dept. of CSE,

SIT Nagpur

Dr. Bhupesh Kumar Dewangan

Associate Professor,

IPR Co-ordinator

DECLARATION

I hereby declare that the project titled "**Project title**" submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.
Name(s) of All Student(s) :

1.**Swanandi Thakre – 25070521086**
2.Girija Mahajan - 25070521087
3.Vansh Gaikwad – 25070521088

Degree: Bachelor of Technology in CSE

Department: Computer Science and Engineering

Title of the project : **TechMart — Shopping Cart Using Linked List**

_____
(Signatures of the All Students)

Date: 11.04.2026

ABSTRACT

TechMart is a browser-based e-commerce shopping cart application that demonstrates the practical implementation of the Singly Linked List data structure in a real-world scenario. The project was built as part of Week 12 of the Hack-O-Week programme, focusing on Data Structures applied to frontend web development.

The application simulates an online tech gadget store where users can browse eight products including wireless earbuds, smart watches, gaming mice, mechanical keyboards, and more. Each product can be added to a cart which is internally managed as a Linked List. Cart operations such as adding items to head or tail, removing items by ID, traversing all nodes, and computing the total price directly correspond to classic Linked List operations.

A real-time visual representation of the linked list structure is rendered on-screen, allowing users to observe how nodes are connected via pointers as items are added or removed. This bridges the gap between theoretical data structure concepts and practical software development.

The project is implemented using HTML5, CSS3, and vanilla JavaScript (ES6+), with no external frameworks, showcasing that sophisticated data structures can be effectively used in client-side web applications.

Attribute	Details
Project Name	TechMart — Shopping Cart Using Linked List
Technology Used	HTML5, CSS3, JavaScript (ES6+)
Data Structure	Singly Linked List
Week	Week 12 — Hack-O-Week
Domain	Frontend Web Development + DSA

TABLE OF CONTENTS

Chapter / Section	Page
Certificate	2
Declaration	3
Abstract	4
Table of Contents	5
Chapter 1: Background and Technical Overview	6-7
Chapter 2: Problem Statement and Motivation	8
Chapter 3: Novelty and Innovative Contributions	9
Chapter 4: Technical Advantages and Practical Usefulness	10
Chapter 5: Detailed Methodology / System Architecture	11-12
Chapter 6: Prior Art and Related Work (Literature Survey)	13
Chapter 7: Applications and Deployment Areas	14
Chapter 8: Conclusion and Future Scope	15
References (IEEE Format)	16
Appendices	17

CHAPTER 1: BACKGROUND AND TECHNICAL OVERVIEW

1.1 Introduction to Linked Lists

A Linked List is one of the most fundamental linear data structures in computer science. Unlike arrays, which store elements in contiguous memory locations, a linked list stores data in a collection of nodes where each node contains a data field and a reference (pointer) to the next node in the sequence.

This structure allows efficient insertion and deletion at any position in the list without reorganizing the entire data structure. Linked Lists form the backbone of many advanced data structures such as stacks, queues, hash tables, and graphs.

1.2 Types of Linked Lists

- Singly Linked List: Each node has one pointer pointing to the next node. The last node points to NULL.
- Doubly Linked List: Each node has two pointers — one to the next node and one to the previous node.
- Circular Linked List: The last node's pointer points back to the head, forming a ring structure.

This project uses a Singly Linked List, implemented in JavaScript, to represent the shopping cart.

1.3 Node Structure

Each node in the TechMart linked list stores a cart item object and a reference to the next node:

Node Property	Description
data.id	Unique product identifier
data.name	Name of the product
data.price	Price in Indian Rupees (₹)
data.quantity	Number of units in cart
data.image	URL of product image
next	Pointer to the next node (null if last)

1.4 Technologies Used

- HTML5 — Semantic structure of the web page
- CSS3 — Styling with CSS Variables, Flexbox, Grid, and animations
- JavaScript ES6+ — Classes, Arrow Functions, Template Literals
- Font Awesome 6.5 — Icon library for UI icons

- Google Fonts — Space Grotesk & Inter font families
- Unsplash API — Real product images loaded via URL

1.5 Project File Structure

File	Purpose
week_12.html	Main HTML structure and layout
12_stylecc.css	All styling, theming, and animations
12_script.js	Linked List implementation and all JavaScript logic

CHAPTER 2: PROBLEM STATEMENT AND MOTIVATION

2.1 Problem Statement

Most e-commerce platforms use abstract array-based or database-driven shopping carts where the underlying data structure is hidden from the developer and not educationally meaningful. There exists a gap between theoretical Data Structure education and real-world application development.

The challenge is to build a fully functional e-commerce shopping cart that explicitly uses a Linked List as its core data structure, making the operations transparent through a live visual representation on screen.

2.2 Motivation

- Data structures are taught theoretically in most curricula without real-world mapping.
- Students struggle to understand how DSA applies to actual software products.
- An interactive, visual demonstration helps solidify understanding of pointer-based structures.
- Frontend web development is one of the most accessible platforms for interactive DSA demos.

2.3 Objectives

1. Implement a Singly Linked List class in JavaScript with full CRUD operations.
2. Build a visually appealing e-commerce UI that interacts with the linked list.
3. Render a real-time graphical visualization of the linked list (HEAD → nodes → NULL).
4. Support add-to-head, add-to-tail, and remove-by-ID linked list operations.
5. Compute cart totals by traversing the linked list.
6. Provide a checkout simulation that clears the list.

2.4 Scope

The project is scoped to a client-side only web application with no backend, database, or server-side logic. All data including products and cart state exists in memory, managed entirely via the linked list data structure.

CHAPTER 3: NOVELTY AND INNOVATIVE CONTRIBUTIONS

3.1 Live Linked List Visualization

The most novel aspect of TechMart is the real-time graphical representation of the linked list inside the cart sidebar. As items are added to the cart, the visualization updates to show:

- HEAD label — indicating the start of the linked list
- Node boxes — showing the first 6 characters of each product name
- Arrow indicators (→) — representing the next pointer between nodes
- NULL label — indicating the end of the list

This transforms an abstract computer science concept into a tangible, interactive visual that users can observe in real time.

3.2 Class-Based OOP Implementation

Unlike most beginner implementations that use functions and arrays, TechMart uses a fully object-oriented approach with two ES6 classes:

- Node class — encapsulates data and the next pointer
- ShoppingCart class — manages head, tail, size, and all list operations

This mirrors how linked lists are taught in Java or C++, but in a browser-native language (JavaScript).

3.3 Dual Insertion Support

TechMart's ShoppingCart class supports both addToHead() and addToTail() methods. While the UI defaults to tail insertion (FIFO order, similar to a real shopping cart), the addToHead() method is available for priority item insertion, demonstrating the flexibility of linked lists over arrays.

3.4 Dark Mode UI with Design System

The CSS is built on a CSS variable-based design system with a professional dark theme matching modern developer tools and tech product aesthetics. The UI includes hover animations, badge labels (Best Seller, New, Sale), image zoom effects, and a sliding cart sidebar with an overlay — all achieved with pure CSS transitions.

CHAPTER 4: TECHNICAL ADVANTAGES AND PRACTICAL USEFULNESS

4.1 Advantages of Linked List over Array for Cart

Feature	Array	Linked List (TechMart)
Insertion at beginning	O(n) — shift elements	O(1) — update head
Insertion at end	O(1) amortized	O(1) — update tail
Deletion by ID	O(n) — splice and re-index	O(n) — traverse, update pointer
Memory allocation	Contiguous block	Dynamic, non-contiguous
Visual pointer concept	Not applicable	Directly visualized

4.2 Educational Value

- Demonstrates Head and Tail pointer management in a practical application.
- Shows traversal algorithms (display, getTotal, getSize) through real data.
- Visual representation bridges textbook diagrams and actual code execution.
- The removeById method demonstrates the pointer re-linking concept visually.

4.3 Performance Characteristics

- addToTail(): O(1) — constant time due to tail pointer maintenance.
- addToHead(): O(1) — constant time due to head pointer maintenance.
- removeById(): O(n) — linear traversal to find the node.
- getTotal(): O(n) — traverses all nodes to sum prices.
- display(): O(n) — traverses all nodes to collect items.
- clear(): O(1) — sets head and tail to null, size to 0.

4.4 Real-World Mapping

This project demonstrates that linked lists are not just academic constructs. Real-world systems such as browser history (forward/back navigation), undo-redo stacks in editors, music playlist management, and process scheduling in operating systems all rely on linked list principles.

CHAPTER 5: DETAILED METHODOLOGY / SYSTEM ARCHITECTURE

5.1 System Architecture Overview

TechMart follows a pure client-side architecture. There is no server, no database, and no API calls (except for loading external fonts and icons). All logic exists within three files communicating through the browser's DOM.

5.2 Linked List Class Design

The ShoppingCart class implements the following methods:

Method	Complexity	Description
addToTail(item)	O(1)	Adds a new node at the end of the list
addToHead(item)	O(1)	Adds a new node at the beginning
removeById(id)	O(n)	Traverses and removes node matching id
display()	O(n)	Returns array of all node data values
getSize()	O(1)	Returns current node count
getTotal()	O(n)	Sums price × quantity for all nodes
clear()	O(1)	Resets head, tail to null and size to 0

5.3 User Interaction Flow

7. Page loads → renderProducts() reads products array and builds product cards in DOM.
8. User clicks Add button → addToCart(productId) creates a cart item object and calls cart.addToTail().
9. updateUI() is called → reads cart.display() and renders all cart items in sidebar.
10. updateLinkedListVisual() → reads cart.display() and builds HEAD → node → node → NULL display.
11. User clicks trash icon → removeFromCart(id) calls cart.removeById(id) and refreshes UI.
12. User clicks Checkout → cart.clear() resets the list, UI updates to empty state.

5.4 CSS Design System

The styling uses CSS Custom Properties (variables) defined in :root, providing a centralized design token system:

- --primary: #6366f1 (Indigo) — buttons and node highlights
- --accent: #22d3ee (Cyan) — prices and HEAD/NULL labels
- --success: #10b981 (Green) — checkout button and toast notifications
- --danger: #ef4444 (Red) — remove buttons and sale badges
- --bg-dark: #020617 — page background
- --bg-card: #0f172a — product cards and sidebar

5.5 Product Data

Eight products are defined as a JavaScript array of objects. Each product has: id, name, desc, price (in INR), image (Unsplash URL), and an optional badge. The product catalogue includes Wireless Earbuds Pro, Smart Watch Ultra, Gaming Mouse RGB, Portable Charger, Mechanical Keyboard, USB-C Hub, Webcam 4K Pro, and Laptop Stand.

CHAPTER 6: PRIOR ART AND RELATED WORK (Literature Survey)

6.1 Linked Lists in Web Development

While linked lists are commonly discussed in academic computer science literature, their direct application in web-based UI projects is relatively uncommon. Most web shopping carts use JavaScript arrays or state management libraries (Redux, Zustand) without exposing the underlying data structure.

6.2 Relevant Academic and Technical References

- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). Introduction to Algorithms (3rd ed.) — provides foundational definitions of linked list operations including traversal, insertion, and deletion.

- Goodrich, M. T., Tamassia, R., & Goldwasser, M. H. (2014). Data Structures and Algorithms in Java — discusses linked list implementation in object-oriented languages, which informed the class-based design of TechMart.

- MDN Web Docs — JavaScript Classes (ES6) — Reference for implementing OOP patterns in modern JavaScript, used for the Node and ShoppingCart class implementations.

- CSS-Tricks (2023). A Complete Guide to CSS Custom Properties — influenced the design token system used in 12_stylecc.css.

6.3 Existing Similar Projects

Several open-source DSA visualization tools exist (e.g., VisuAlgo, CS50's tools), but they are standalone educational tools not integrated with actual product interaction. TechMart differs by combining both — a real product browsing experience with embedded, functional DSA operations.

CHAPTER 7: APPLICATIONS AND DEPLOYMENT AREAS

7.1 Direct Applications

- Educational DSA Demonstrations: Can be used by faculty to visually explain linked lists in class.
- Coding Bootcamps: Ideal as a Week-level project in frontend + DSA integrated courses.
- Portfolio Projects: Demonstrates both UI/UX skill and computer science fundamentals.
- Interview Preparation: Reinforces understanding of pointer manipulation and traversal.

7.2 Extended Deployment Possibilities

- With a backend (Node.js + MongoDB), TechMart can become a real e-commerce store.
- The linked list cart can be replaced with a doubly linked list to support undo/redo cart changes.
- A priority queue variant could allow premium items to always appear at the head of the cart.
- The visualization module can be reused for other DSA topics like stacks, queues, and trees.

7.3 Industry Relevance

The project is directly relevant to the following industry domains:

Domain	Relevance
E-commerce	Cart management, order processing queues
EdTech	Interactive DSA learning tools
Frontend Engineering	DOM manipulation, event-driven programming
System Design	Understanding memory management and pointers

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

TechMart successfully demonstrates the real-world application of a Singly Linked List data structure within a modern, visually engaging e-commerce web interface. The project bridges the gap between theoretical DSA concepts taught in academia and practical frontend web development.

Through this project, the following were achieved:

- A fully functional ES6 class-based Linked List implementation in JavaScript.
- Complete cart operations including add-to-tail, add-to-head, remove-by-ID, traversal, and total calculation.
- A real-time visual linked list representation (HEAD → nodes → NULL) inside the cart sidebar.
- A professional dark-theme UI with responsive design, CSS design tokens, and smooth animations.

This project proves that data structures are not isolated academic constructs — they power real products, and understanding them deeply makes a developer significantly more capable.

8.2 Future Scope

- Doubly Linked List: Add a previous pointer to each node to support backward traversal and undo cart operations.
- Quantity Management: Allow incrementing/decrementing quantity of an existing item instead of adding duplicate nodes.
- Persistent Storage: Use localStorage or IndexedDB to persist cart state across browser sessions.
- Backend Integration: Connect to a Node.js/Express REST API with MongoDB for real product and order management.
- Sorting by Price: Implement merge sort or insertion sort on the linked list to sort cart items by price.
- Search Functionality: Add a product search bar using linear search through the products array.
- Multiple Data Structures Tab: Extend the visualizer to demonstrate stacks, queues, and binary trees using the same product data.

REFERENCES (IEEE Format)

13. [1] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, 3rd ed. Cambridge, MA: MIT Press, 2009.

14. [2] M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, Data Structures and Algorithms in Java, 6th ed. Hoboken, NJ: Wiley, 2014.

15. [3] MDN Web Docs, "JavaScript Classes," Mozilla Foundation. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Classes. [Accessed: Apr. 2026].

16. [4] MDN Web Docs, "Linked List," Mozilla Foundation. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/JavaScript. [Accessed: Apr. 2026].

17. [5] CSS-Tricks, "A Complete Guide to Custom Properties," 2023. [Online]. Available: https://css-tricks.com/a-complete-guide-to-custom-properties/. [Accessed: Apr. 2026].

18. [6] Font Awesome, "Font Awesome 6 Icons," 2024. [Online]. Available: https://fontawesome.com. [Accessed: Apr. 2026].

19. [7] Google Fonts, "Space Grotesk & Inter Typefaces," Google LLC. [Online]. Available: https://fonts.google.com. [Accessed: Apr. 2026].

APPENDICES

Appendix A: Key Code Snippet — ShoppingCart Class

The following is the core JavaScript implementation of the Singly Linked List used as the shopping cart:

```
class Node { constructor(data) { this.data = data; this.next = null; } }
class ShoppingCart { constructor() { this.head = null; this.tail = null; this.size = 0; } }
```

Appendix B: Product Catalogue

ID	Product Name	Description	Price (₹)	Badge
1	Wireless Earbuds Pro	Active noise cancellation	₹8,990	Best Seller
2	Smart Watch Ultra	Health & fitness tracking	₹15,990	New
3	Gaming Mouse RGB	16000 DPI precision	₹2,490	—
4	Portable Charger 20000mAh	Fast charging support	₹1,990	Sale
5	Mechanical Keyboard	RGB backlit switches	₹4,990	—
6	USB-C Hub 7-in-1	All ports in one	₹2,990	—
7	Webcam 4K Pro	Auto-focus & HDR	₹7,990	Popular
8	Laptop Stand	Ergonomic aluminum	₹1,490	—

Appendix C: Browser Compatibility

- Google Chrome 90+ — Fully supported
- Mozilla Firefox 88+ — Fully supported
- Microsoft Edge 90+ — Fully supported
- Safari 14+ — Fully supported
- Mobile browsers (Chrome/Safari on Android/iOS) — Responsive layout supported